Linear maps

Let V and W be vector spaces over a field F.

Theorem 1 (Rank–nullity)**.** *For a linear map $T\colon V \to W$ with V finite-dimensional,*

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

A change of basis acts on coordinates by a matrix:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \tag{1}$$

$$\det R_\theta = \cos^2\theta + \sin^2\theta = 1. \tag{2}$$